

Hierarchy Problem Talk



"The newest numbers just came in, we're gonna have to fine tune some more..."

Standard Model

Gauge Group:

$$\underset{\substack{\uparrow \\ \text{QCD}}}{\text{SU}(3)_c} \times \underbrace{\text{SU}(2)_L \times \text{U}(1)}_{\text{Electroweak}}$$

Matter: we write how it couples to the gauge fields in the following way: $(c, L)_Y$

$$Q : (3, 2)_{Y_Q}$$

$$U : (\bar{3}, 1)_{-2/3}$$

$$D : (\bar{3}, 1)_{Y_D}$$

$$L : (1, 2)_{Y_L}$$

$$E : (1, 1)_1$$

$$H : (1, 2)_{Y_H}$$

quarks live in fundamental rep of $\text{SU}(3)_c$ with the three dimensional space being called "color"

the right handed and left handed u and d transform independently w.r.t CP violation

Lagrangian:

$$\mathcal{L} = \mathcal{L}_{\text{kin}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yuk}}$$

$$\mathcal{L}_{\text{Higgs}} = \mu^2 |\phi|^2 - \lambda |\phi|^4 \quad \text{where } \phi \text{ is a vector in } (1, 2)_{Y_H}$$

$$\mathcal{L}_{\text{Yuk}} = y_{ij}^e \bar{L}^i \phi E^j + y_{ij}^d \bar{Q}^i \phi D^j + y_{ij}^u \bar{Q}^i \tilde{\phi} U^j + \text{h.c.}$$

↑
indices for the generations!!

Hidden inside $\mathcal{L}_{\text{kin}}$ is the fact that the SM is CP violating, which translates to the fact the left and right handed fermions live in different reps. For example, the kinetic term for L and E are:

$$i \bar{L}_i (\not{\partial} - i g_W \tau_a - i g_Y Y_L \not{B}) L_i \quad \leftarrow \text{couples to } W, W \text{ being the } \text{SU}(2)_L \text{ gauge field.}$$

$$i \bar{e}_R (\not{\partial} - i g_Y Y_e \not{B}) e_R \quad \leftarrow W \text{ neutral}$$

New Solns To Hierarchy Problem

New Global Symmetries

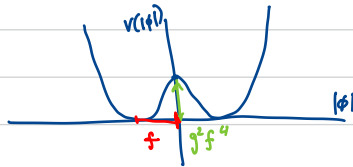
Basic idea will be to treat Higgs as a Goldstone boson. Plays well with naturalness:

Higgs is Goldstone $\rightarrow$ mass = 0

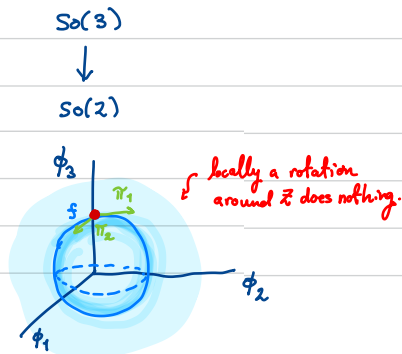
add mass term $\Rightarrow$ new sym when mass taken to zero: $\Rightarrow$ Technical Naturalness:
the Goldstone shift symmetry!
Mass corrections are not quadratically divergent

As a toy model to understand the procedure for the global symmetry route, let's consider a toy model of $SO(3)$ in the fundamental rep, with Lagrangian:

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{g^2}{8} (|\phi|^2 - f^2)^2$$



Now we must pick a vacuum state to expand around. let's say we choose $\begin{pmatrix} 0 \\ f \end{pmatrix}$. Pictorially:



We can furnish all of field space by including a radial coordinate to get:

$$\vec{\Phi} = \exp\left(i \frac{\sigma}{f} \pi(x) \cdot T_i\right) \begin{pmatrix} 0 \\ 0 \\ f + \sigma(x) \end{pmatrix} \quad \text{w/} \quad T_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & i \\ 0 & -i & 0 \end{pmatrix} \quad \text{and} \quad T_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & i \\ -i & 0 & 0 \end{pmatrix}$$

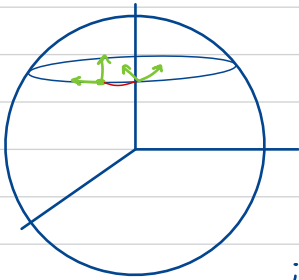
One can plug this form of $\vec{\Phi}$ into the Lagrangian to get the following:

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \sigma \partial^\mu \sigma) - \left(\frac{g^2}{2} \right) \sigma^2 - \frac{g^2 f}{2} \sigma^3 + \dots \\ + \frac{1}{2} \left(1 + \frac{\sigma^2}{f^2} \right) \left[\partial_\mu \pi^T \partial^\mu \pi - \frac{1}{8} \frac{\pi^2}{f^2} \partial_\mu \pi^T \partial^\mu \pi + \dots \right]$$

But now if we want normal Higgs pheno, The Higgs need to break some symmetry, but where is the potential coming from?

$$\Phi^\dagger \Phi \sim (f + \sigma)^2$$

Answer: gauge the unbroken symmetry. Where is our $SO(2)$ symmetry? This manifests in our goldstone modes linearly.



$$\leadsto \Phi \rightarrow e^{i\sqrt{2}\alpha \cdot T} \Phi \Leftrightarrow \vec{\pi} = e^{\alpha \cdot (-i)} \vec{\pi}$$

$\Downarrow$ gauge

$$\partial_\mu \Phi \rightarrow D_\mu \Phi = \left(\partial_\mu - i\sqrt{2} e A_\mu T \right) \Phi$$

$\Downarrow$

if we write $H = \frac{\pi_1 - i\pi_2}{\sqrt{2}}$ we get the Lagrangian:

$$\frac{1}{2} \left(1 + \frac{\sigma^2}{f^2} \right) \cdot \left\{ \frac{f^2}{4|H|^2} \sin^2\left(\frac{\sqrt{2}H}{f}\right) D_\mu H D^\mu H + \frac{f^2}{4|H|^4} \left(2 \frac{|H|^2}{f^2} - \sin^2\left(\frac{\sqrt{2}H}{f}\right) \right) (\partial_\mu |H|^2)^2 \right\}$$

Since we explicitly broke the $SO(3) \rightarrow U(1)$, we'll get radiative corrections to m_H^2 . Then $\langle H \rangle$ can just break that potential.

What have we accomplished? Remember our (technical) naturalness arguments:

$$m_H^2 = \underbrace{c\Lambda^2}_{\text{tree level}} + \underbrace{S m_H^2}_{\text{radiative corrections}}$$

Radiative corrections will come from terms that break the symmetry. Brief aside: how to make predictions with naturalness. Ex: Splitting of pion masses. QCD says pions should be massless as they are goldstones of Chiral symmetry breaking. However we have terms in the SM that break that sym: QED. In particular $\pi^\pm$ are charged so couple to $U(1)_{EM}$, but π^0 does not.

$$m_{\pi^\pm}^2 \sim \cancel{c\Lambda^2} + S m_{\pi^\pm}^2$$



$$m_{\pi^\pm}^2 - m_{\pi^0}^2 \sim (355 \text{ MeV})^2 \sim \frac{3e^2}{16\pi^2} \Lambda^2 \Rightarrow \text{New physics at } \Lambda \sim 850 \text{ MeV}$$

Low and behold we find the ρ meson at $\sim 775 \text{ MeV}$

Going back to our model, we do a similar argument but in reverse: none of the SM respects our $SO(3)$ sym, so we get corrections to m_H :

$$m_H^2 = \cancel{c\Lambda^2} + S m_H^2 \sim \frac{\Lambda^2}{16\pi^2} (-6y_t + \dots)$$

$\Downarrow$
 $m_H^2 \sim \frac{y_t^2}{16\pi^2} \Lambda^2$

from a $t\bar{t} H A$ yukawa

Now we should see new physics at $\Lambda^2 \sim \frac{16\pi^2}{y_t^2} m_H^2$. But we don't see that! So if we run the argument in reverse: Λ from exp is $\sim 5 \text{ TeV}$, so to match small m_H^2 , we need fine tuning.

You could try to argue that we should've made the SM $SO(3)$ symmetric:

$$\mathcal{L} \supset -y_t t_R H Q$$

↓

$$\mathcal{L} \supset -y_t (t_R + T_R) \Phi \tilde{Q} \quad \leftarrow \begin{pmatrix} Q \\ T_L \end{pmatrix}$$

↓ $SO(3) \rightarrow SO(2)$ sym breaking.

$$\mathcal{L} \supset -y_t t_R H t_L + \frac{y_t^2}{m_T} H^\dagger H T_R^\dagger T_L \quad \text{w/ } m_T \sim y_t f$$



This gets rid of our fine tuning problem:

$$\delta m_t^2 = -\frac{3y_t^2}{4\pi^2} m_T^2 \ln\left(\frac{\Lambda}{m_T}\right)$$

But we just don't see m_T ... Can see similar issues with $SU(3) \rightarrow SU(2)$.

Neutral Naturalness

H is not a goldstone. Start with two standard models and a $\mathbb{Z}_2$ sym that swaps them

$$SM_A \longleftrightarrow SM_B$$

+

Higgs potential that
lets them talk

Upshot we'll end up with similar corrections as above but with $m_\tau \rightarrow$ particle in SM_B .

$$SU(4) \rightarrow SU(3)$$

$15 - 8 = 7$ generators $\Rightarrow$ ^{says} 3 goldstones eaten by the $SU(2)$ in $SM_B \Rightarrow 7 - 3 = 4$ goldstones.

$$H = \begin{pmatrix} H_A \\ H_B \end{pmatrix} = \exp \left(\frac{i}{f} \begin{pmatrix} 0 & \begin{matrix} h_1 \\ h_2 \\ 0 \end{matrix} \\ \begin{matrix} h_1 h_2 \\ 0 \end{matrix} & 0 \end{pmatrix} \right) \begin{pmatrix} 0 \\ 0 \\ 0 \\ f \end{pmatrix} = \begin{pmatrix} \vec{h} + \frac{h_1^2}{6f^2} + \dots \\ 0 \\ f - \frac{1}{2} \frac{h_1^2}{f} + \dots \end{pmatrix}$$

Therefore our Yukawa couplings become

$$\lambda H_A Q_A t_A + \lambda H_B Q_B t_B$$

$\downarrow$

$$\lambda h Q_A t_A + \lambda \left(f - \frac{h_1^2}{2f} \right) Q_B t_B$$

$\downarrow$



$$-\frac{6\lambda^2}{16\pi^2} \Lambda^2 + \frac{6\lambda^2}{16\pi^2} \Lambda^2 - \frac{3\lambda^2}{4\pi^2} \frac{m_B^2}{\Lambda^2} \log \left(\frac{\Lambda}{m_B} \right)$$

Why do these help us? We still want $\Lambda \sim M_{\text{Pl}}$, but our colliders just don't directly probe these SM particles, so we can't put constraints on Λ . Now this sounds like cheating. But in fact this model is still predictive.

- ① Higgs kinetic term deviations
- ② Decays into SMs

① Recall that we had a kinetic term of the following form:

$$|\partial_\mu H|^2 \sim |\partial_\mu h|^2 + \frac{h^\dagger h |\partial_\mu H|^2}{\Lambda^2}$$

This is the primary signature of this theory.

② The Higgs is also still coupled to the fermions of SM, so it could possibly decay into SMs.

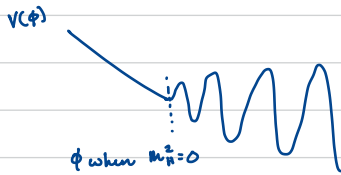
Relaxion

Say we have a field ϕ with a shift sym except w/ coupling $g\phi|H|^2$.

$$(-M^2 + g\phi)|H|^2 + V(g\phi) + \frac{1}{32\pi^2} \frac{\phi}{f} G\tilde{G}$$

High level: track semi-classics of $\langle\phi\rangle$, if $\langle\phi\rangle \ll 0$ to $\langle\phi\rangle \gg 0$, we go from no QCD particle masses to having masses.

This turns out to increase the bumps of the potential



$$\Leftrightarrow \text{wiggles: } \langle G\tilde{G} \rangle \sim \Lambda_{\text{QCD}}^4 \cos(\phi/f)$$

$$\Lambda^4 \sim f_\pi^2 m_\pi^2 \sim f_\pi^2 \cdot (f_\pi v \gamma)$$

When do you stop · force balance

$$g \sim \frac{\gamma v f_\pi^3}{M^2 f}$$

$$\left(\begin{array}{c} \text{SU}(3) \rightarrow \text{SU}(2) \\ \text{reference} \end{array} \right)$$

Let's see how this is borne out w/ $\text{SU}(3) \rightarrow \text{SU}(2)$ sym breaking. Say we have a field ϕ be a vector in the fundamental rep of $\text{SU}(3)$. Let's say we break ϕ by giving it a VEV:

$$\phi = \begin{pmatrix} 0 \\ 0 \\ f \end{pmatrix}$$

$\nwarrow$ expansions around this are the massive modes
 $\swarrow$ particular state is vacuum manifold
 $\downarrow$

$$\phi = \exp \left(\frac{i}{f} \begin{pmatrix} -\frac{\gamma_2}{2} & 0 & H_1 \\ 0 & -\frac{\gamma_2}{2} & H_2 \\ H_1^\dagger & H_2^\dagger & \gamma_2 \end{pmatrix} \right) \begin{pmatrix} 0 \\ 0 \\ f \end{pmatrix}$$

γ_2 traceless Hermitian matrices

Now we can calculate our kinetic term:

$$\mathcal{L}_{\text{kin}} = f^2 |\partial_\mu \phi|^2 = |\partial_\mu H|^2 + \frac{H^\dagger H}{f^2} |\partial_\mu H|^2$$

$$\mathcal{L}_{\text{Yukawa}} = -\lambda_1 (\tilde{t}_R^\dagger + \lambda_2 \tilde{\tau}_R^\dagger) \phi^\dagger Q_3 + \text{h.c.}$$

$$\sim -\lambda_1 t_R^\dagger \tilde{H} Q_3 - \lambda_1 \tau_R^\dagger \tilde{H} Q_3 + \frac{\lambda_1^2}{m_\tau} (H^\dagger H) \tau_R^\dagger T_L + \text{h.c.} + \dots$$

$$\begin{array}{ccc} \cdots \bigcirc \cdots & \cdots \bigcirc \cdots & \bigcirc \\ \hline -\frac{6\lambda_1^2}{16\pi^2} \Lambda^2 & -\frac{6\lambda_1^2}{16\pi^2} \Lambda^2 & + \frac{6(\lambda_1^2 + \lambda_\tau^2)}{16\pi^2} \Lambda^2 + \text{finite} \end{array}$$